Enhancing Decision Making

Chapter 12 — Full Depth Study Notes

1. Enhancing Decision Making

> Core Idea: Improving the quality of decisions by using information systems.

In an organization, decisions are made every day. Some are simple, some are complex.

Without proper information:

- Decisions become guesses
- Mistakes increase
- Costs rise

Information systems improve decision making by providing:

- Accuracy — reliable, clean data
- Speed — faster access to information
- Reliability — consistent, repeatable results

2. Decision Making

> Definition: Choosing the best option from different alternatives.

For a decision to exist, three conditions must be met:

1. A problem must exist
2. Multiple solutions must be available
3. One solution must be selected

Example

Problem: Sales are decreasing.

Options:

- Increase marketing spend
- Reduce price
- Improve the product

Decision: Select the best option based on available data and analysis.

3. Types of Decisions

1. Structured Decisions

Routine, repetitive decisions where the rules are already clearly defined.

- Salary calculation
- Order processing
- Invoice generation

These can be fully automated by information systems.

2. Semi-Structured Decisions

Decisions where some rules exist but human judgment is still required.

- Budget planning
- Sales forecasting
- Hiring decisions

System helps with data and analysis. Human makes the final call.

3. Unstructured Decisions

Complex, unique situations with no predefined rules.

- Entering a new market
- Setting long-term business strategy
- Crisis response

Mostly human-driven. System only provides supporting information.

Decision Type	Nature	Example	System Role
Structured	Routine, rule-based	Salary calculation	Full automation
Semi-Structured	Partial rules + judgment	Budget planning	Support & analysis
Unstructured	Complex, no fixed rules	Market entry	Information only

4. Decision Making Process

Every decision, regardless of type, follows these four steps:

1. Identify the Problem — Understand what is wrong and why
2. Develop Alternatives — Find all possible solutions
3. Select Best Option — Evaluate and choose the best solution
4. Implement the Decision — Apply it and monitor the result

> Important Insight: A correct decision depends on correct problem identification. If Step 1 is wrong, every step after it will also be wrong.

5. How Information Systems Support Decision Making

Information systems support every step of the decision-making process by:

- Providing accurate, real-time data
- Processing large volumes of data quickly
- Generating structured reports for managers
- Supporting what-if analysis and scenarios

Example

Without system: Manager guesses demand based on experience alone.

With system: Manager uses historical data, trend reports, and forecasting models to decide accurately.

6. Decisions Vary by Level

Different organizational levels face different types of decisions.

Operational Level

- Routine, day-to-day decisions
- Fully defined, structured
- Example: Approving a transaction, recording attendance

Managerial Level

- Medium-term planning decisions
- Semi-structured, need some judgment
- Example: Monthly budget allocation, performance review

Strategic Level

- Long-term, high-impact decisions
- Unstructured, complex
- Example: Entering a new market, acquiring a company

> Rule: Higher level □ More complex decisions □ More judgment required.

7. Decisions Vary by Structure

Decision type maps directly to organizational level:

Decision Structure	Organizational Level	Information Need
Structured	Operational	Detailed, real-time transaction data
Semi-Structured	Managerial	Summarized reports, trends
Unstructured	Strategic	External data, forecasts, insights

> Key Connection: This is why different systems exist for different levels.
> TPS □ Operational • MIS/DSS □ Managerial • ESS □ Strategic

8. Business Intelligence Systems

> Definition: Systems used to collect, process, and analyze data for better decision making.

Business Intelligence is not one single system. It is a collection of systems, each serving a different function.

The flow:

1. Data collected from all business sources
2. Data processed and cleaned
3. Information generated through analysis
4. Decisions made based on insights

9. Types of Business Intelligence Systems

1. Management Information System (MIS)

Provides structured reports to support managerial decisions.

- Generates reports from operational data
- Used by middle managers
- Helps in planning and performance monitoring

2. Marketing Information System

Supports marketing decisions using internal and external data.

Uses data like:

- Sales figures and history
- Past trends and seasonal patterns
- Government policies and market conditions

Helps in:

- Identifying key customers and segments
- Finding primary revenue sources
- Planning promotions and campaigns

Examples from notes:

- Discount strategies
- After-sale service planning
- New product offerings
- Order prioritization decisions

3. Human Resource Information System (HRIS)

Manages all employee-related data and HR processes. Divided into four modules:

a. HR Management Module

Stores comprehensive employee records.

- Employee personal data and history
- Skills, qualifications, and certifications
- Training completed and achievements

Used for:

- Recruitment and selection
- Promotion decisions
- Compensation planning

b. Training Module

Tracks and manages all employee training activities.

- Records training history per employee
- Supports the organization as a learning entity
- Suggests relevant courses based on role and gaps
- Assists in training budget decisions

c. Payroll Module

Handles all aspects of employee compensation.

- Calculates base salary and overtime
- Processes tax deductions and benefits
- Manages incentives and bonuses
- Tracks attendance for pay calculation

> Important: The Payroll Module is directly linked with the financial system.
> Any payroll change automatically updates financial records.

d. Time & Attendance Module

Tracks employee working hours and efficiency.

- Records clock-in and clock-out times
- Measures individual and team efficiency
- Feeds data directly into the Payroll Module

10. Final Understanding

Chapter 12 covers the following topics in order:

1. Decision making — meaning and definition
2. Types of decisions — structured, semi-structured, unstructured
3. Decision making process — four steps
4. Decision levels — operational, managerial, strategic
5. Decision structure — how type maps to level
6. Role of information systems in supporting decisions
7. Business Intelligence Systems — definition and flow
8. Marketing Information System — purpose, data, outputs

9. HRIS and its four modules — HR, Training, Payroll, Attendance

> Remember: Business Intelligence is not one system — it is a collection.
>
> Each system targets a specific function: MIS for management, Marketing IS for campaigns, HRIS for people management.